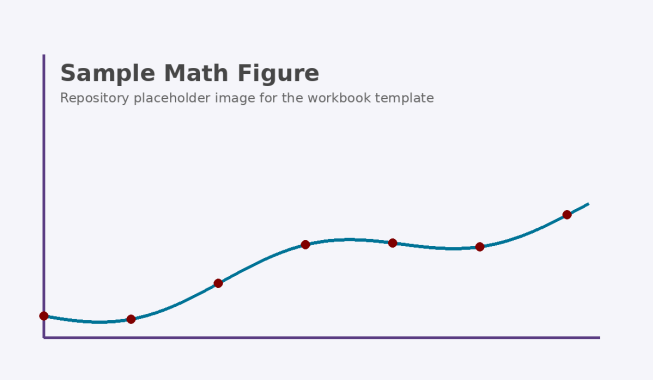
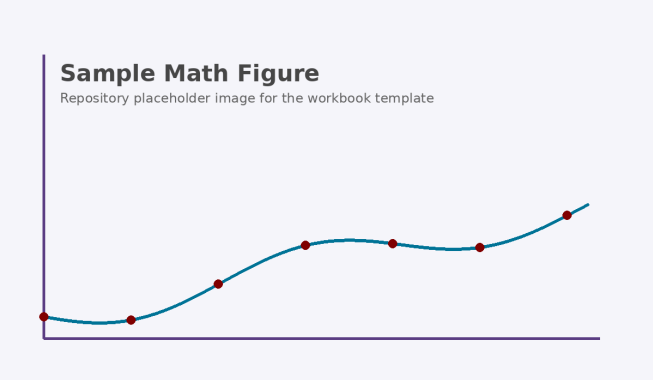


Sample Math Workbook Template

MATH 16300 Lecture notes

Subhadip Chowdhury, PhD



Spring 2026

A Repository-Style Minimal Working Example

This page is inserted with `\PreTOCMatter`. It is useful for a short note, dedication, or instructor-facing cover page before the table of contents.

Mathematics, rightly viewed, possesses not only truth, but supreme beauty.

— Bertrand Russell

How to Use This Template iv

1 Introduction 1
1.1 Repository structure . . . 1

2 Theorem-Like Environments and Workbook Features 3
2.1 A few sample statements . . . 3
2.2 Exercises and solutions . . . 4

3 Macro Sampler 5
3.1 Basic notation . . . 5
3.2 Delimiters, arrows, and elementary symbols . . . 5
3.3 Derivatives, differentials, and integrals . . . 5
3.4 Algebra, transforms, and special symbols . . . 5
3.5 Operator macros . . . 6
3.6 Inline answer toggles and spacing . . . 6
3.7 More vector notation . . . 6

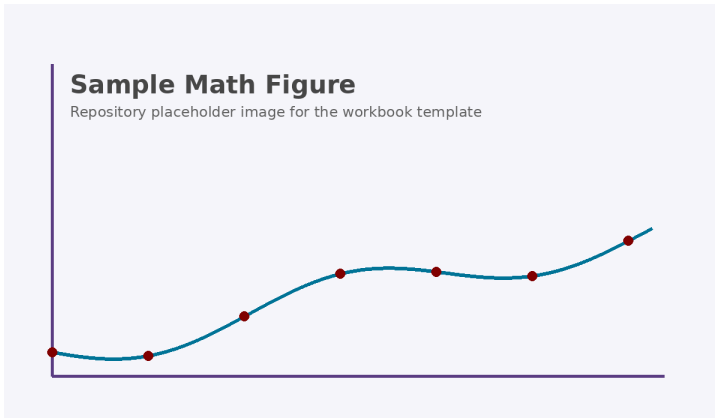
A Appendix: Cross-References and Citations 7

B Appendix: Worksheet Files in the Repository 8

This repository demonstrates workbook mode in `main.tex` and worksheet mode in `assignments/assignment1.tex` and `assignments/assignment2.tex`. The workbook includes chapter files from `chapters/` and appendix files from `appendices/`. It also cites a few sample references such as [**apostol-calculus**, **axler-linear**] and uses a placeholder image stored at `images/sampleimage.png`.

Module 1 | Introduction

This introduction shows the workbook voice and layout. The class provides convenient color-switching macros such as blue text, maroon text, and lake-colored text. It also provides an emphasis macro: **important ideas can be highlighted consistently**. Throughout the template, cross-references use `\cref`, for example theorem 3 and definition 1.



§1.1 Repository structure

The top-level file `main.tex` compiles the workbook. The folder `chapters/` stores modular chapter files, `appendices/` stores appendices, and `assignments/` stores standalone worksheet-mode assignments. The command `\blankfill` is useful for fill-in-the-blank prompts: Course website _____

Template note: This note box shows one recommended use of the class: keep exposition in workbook mode and use the worksheet option for short assignments or handouts.

Compilation reminder: Because the class uses `biblatex` with the `biber` backend, the workbook should be compiled with a LaTeX → Biber → LaTeX → LaTeX workflow.

DIGRESSION

A compact workbook often benefits from short historical or conceptual side paths. This environment is a natural place to discuss why a result matters before returning to the main thread.

LECTURE PLAN – OPENING MODULE

Learning Objectives

Students should recognize the repository layout, identify the main entry points, and understand how the class structures theorem-like environments.

Timing (10 minutes)

Spend a few minutes walking through the folder structure and the sample chapter files.

Instructor note: The lecture-plan and staff-note environments are most useful in instructor-facing copies of the workbook. Set `\showplanfalse` in a student-facing version if you want to suppress them.

Module 2 | Theorem-Like Environments and Workbook Features

§2.1 A few sample statements

Definition 2.1.1: Inner product space

An *inner product space* is a vector space V over $\mathbb{R}$ or $\mathbb{C}$ equipped with a function $\langle \cdot, \cdot \rangle : \mathrm{V} \times \mathrm{V} \to \mathbb{R}$ or $\mathbb{C}$ satisfying the usual axioms.

Axiom 2.1.2: Order axiom

For every $x \in \mathbb{R}$, exactly one of the following holds: $x < 0$, $x = 0$, or $x > 0$.

Theorem 2.1.3: Power rule

If $f(x) = x^n$ for $n \in \mathbb{N}$, then $\dfrac{\mathrm{d}}{\mathrm{d}x} f(x) = nx^{n-1}$.

Proof of theorem 3.

One proves this using a binomial theorem identity. The displayed derivative notation in this class can be written as $\dfrac{\mathrm{d}}{\mathrm{d}x}(x^n) = nx^{n-1}$. ■

Proposition 2.1.4: A norm identity

For vectors $\vec{u}, \vec{v} \in \mathrm{V}$, the quantity $\|\vec{u} + \vec{v}\|$ depends continuously on $\vec{u}$ and $\vec{v}$.

Claim 2.1.5: A simple estimate

If $|x| \leq 1$, then $|x^2| \leq 1$.

Example 2.1.6: A derivative computation

If $f(x) = x^3$, then $\dfrac{\mathrm{d}f}{\mathrm{d}x} = 3x^2$ and $f(x)\Big|_0^1 = 1$.

§2.2 Exercises and solutions

Question 1. Warm-up

Compute $\int_0^1 x^2\,\mathrm{d}x$ *and compare your answer with the statement of theorem 3.*

Solution. We have $\int_0^1 x^2\,\mathrm{d}x = \left.\frac{x^3}{3}\right|_0^1 = \quad .$ ■

Question 2. Vectors

Let $\vec{\mathbf{R}}(t) = \langle t, t^2, t^3\rangle$. *Write down* $\frac{\mathrm{d}\vec{\mathbf{R}}}{\mathrm{d}t}$, $\frac{\mathrm{d}x}{\mathrm{d}t}$, $\frac{\mathrm{d}y}{\mathrm{d}t}$, *and* $\frac{\mathrm{d}z}{\mathrm{d}t}$.

Solution. $\frac{\mathrm{d}\vec{\mathbf{R}}}{\mathrm{d}t} = \langle 1, 2t, 3t^2\rangle$, while $\frac{\mathrm{d}x}{\mathrm{d}t} = 1$, $\frac{\mathrm{d}y}{\mathrm{d}t} = 2t$, and $\frac{\mathrm{d}z}{\mathrm{d}t} = 3t^2$. ■

A typical proof skeleton might list a proof sketch or an algorithm.

CODE:

```
import math

def midpoint_rule(f, a, b):
    m = 0.5 * (a + b)
    return (b - a) * f(m)

print(midpoint_rule(lambda x: x**2, 0.0, 1.0))
```

Module 3 | Macro Sampler

§3.1 Basic notation

The class provides the common number systems $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$ and also a generic field $\mathbb{F}$. It also includes calligraphic helpers such as $\mathcal{U}$, $\mathcal{P}(\mathbb{N})$, and $\mathcal{O}(n^2)$, as well as a Lie-theoretic letter $\mathfrak{u}$ and an affine space symbol $\mathbf{A}^2$.

Write your name here: and a longer response here: _____

§3.2 Delimiters, arrows, and elementary symbols

$$\langle u, v\rangle = u_1 v_1 + u_2 v_2, \qquad \left\lceil \frac{1}{7} \right\rceil = 1, \qquad \left\lfloor \frac{1}{7} \right\rfloor = 0, \qquad x_n \rightharpoonup x, \qquad \mu_n \overset{*}{\rightharpoonup} \mu.$$

$$\mathrm{A} = \{x \in \mathbb{R} \mid x^2 \le 1\}, \qquad \varphi(x) = x^2 + \varepsilon, \qquad \operatorname{cis}\theta = \cos\theta + i\sin\theta, \qquad \nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right).$$

$$\overline{\mathrm{A}} = [-1, 1], \qquad f_\varepsilon(x) = x^2 + \varepsilon.$$

§3.3 Derivatives, differentials, and integrals

$$\frac{\mathrm{d}}{\mathrm{d}x}(x^3) = 3x^2, \qquad \frac{\mathrm{d}}{\mathrm{d}y}(xy) = x, \qquad \frac{\mathrm{d}}{\mathrm{d}t}(t^4) = 4t^3.$$

$$\int_0^1 x^2\,\mathrm{d}x, \qquad \oint_{|z|=1} \frac{1}{z}\,\mathrm{d}z, \qquad \iint_{[0,1]^2} xy\,\mathrm{d}\mathrm{A}, \qquad \iiint_{[0,1]^3} xyz\,\mathrm{d}\mathrm{V}.$$

$$\lim_{n\to\infty} \sum_{k=1}^{n} \frac{1}{n} = 1, \qquad \sum_{k=1}^{n} k = \frac{n(n+1)}{2}.$$

$$\frac{\mathrm{d}f}{\mathrm{d}x},\ \mathrm{d}r,\ \mathrm{d}u,\ \mathrm{d}v,\ \mathrm{d}y,\ \mathrm{d}s,\ \mathrm{d}\theta,\ \mathrm{d}\sigma, \qquad \frac{\mathrm{d}y}{\mathrm{d}x},\ \frac{\mathrm{d}x}{\mathrm{d}t},\ \frac{\mathrm{d}y}{\mathrm{d}t},\ \frac{\mathrm{d}z}{\mathrm{d}t},\ \frac{\mathrm{d}\mathrm{P}}{\mathrm{d}t}.$$

§3.4 Algebra, transforms, and special symbols

$$\mathbb{Z}[\sqrt{2}], \qquad \mathbb{Q}(\sqrt{5}), \qquad \mathscr{L}[\sin t] = \frac{1}{s^2+1}, \qquad \mathscr{L}^{-1}\left[\frac{1}{s^2+1}\right] = \sin t.$$

$$\frac{e^x - 1}{x} \overset{\text{L}}{\overset{\downarrow}{=}} 1 \quad \text{as } x \to 0.$$

§3.5 Operator macros

$\mathrm{tr}(A)$, $\mathrm{Int}([0,1])$, $\mathrm{arcsec}(2)$, $\mathrm{arccot}(1)$, $\mathrm{arccsc}(2)$, $\mathrm{charpoly}(A)$, $\mathrm{lcm}(12,18)$.

$\mathrm{Dom}(T)$, $\mathrm{Rng}(T)$, $\mathrm{Codom}(T)$, $\mathrm{Col}(A)$, $\mathrm{Nul}(A)$, $\mathrm{rank}(A)$, $\mathrm{span}\{v_1, v_2\}$, $\mathrm{proj}_W(v)$, $\mathrm{Row}(A)$.

$\mathrm{erf}(1)$, $\mathrm{diag}(1,2,3)$, $\mathrm{Proj}(S)$, $\mathrm{curl}\,\vec{\mathbf{F}}$, $\mathrm{div}\,\vec{\mathbf{F}}$.

§3.6 Inline answer toggles and spacing

The inline solution macro can be used inside prose: the value of $\int_0^1 x\,\mathrm{d}x$ is .

This paragraph appears after `\myskip`.

This paragraph appears after `\mysskip`.

This paragraph appears after `\myssskip`.

§3.7 More vector notation

Let $\vec{v} = \langle 1, 2\rangle$ and $\hat{\boldsymbol{e}}_1 = \langle 1, 0\rangle$. The reciprocal notation $\frac{1}{n}$ can also be combined with vector expressions, for example $\frac{1}{2}\vec{v}$.

Module A | Appendix: Cross-References and Citations

This appendix exists mainly to show that the workbook can cite references from a shared `references.bib` file. For example, a standard calculus text is [**apostol-calculus**], while a clean linear algebra text is [**axler-linear**].

> **Cross-reference example:** Earlier we introduced definition 1, proved theorem 3, and used example 6 as a worked example.

Question 3. **Reflection**

Write two sentences explaining how you would adapt this repository for a quarter-long course.

Module B | Appendix: Worksheet Files in the Repository

The files `assignments/assignment1.tex` and `assignments/assignment2.tex` are standalone worksheet documents using the `worksheet` option. They show how to keep handouts separate from `main.tex` while still sharing the same class file and macro vocabulary.

GitHub tip: When you upload this repository, it is helpful to include a short README that explains which files compile independently and which files are included only through `main.tex`.

Digression

A modular folder structure makes it much easier to reuse notation, styling, and reference data across a whole sequence of lecture notes and assignments.